
Advanced Programming

ÁRPÁD TAMÁSI · ELTE FACULTY OF SOCIAL SCIENCES

SESSION 2 — 15 SEPTEMBER 2026

How the grade is made

Homework*

40% — due Friday 20:00

Final demo

40% — at the last two sessions, **your classmates are the jury**

Instructor

20% — my own assessment

***Late:** by Monday 20:00 it counts **half** · after that, **not at all**

*The full rules: the **Requirements** page on Canvas.*

Today's session

	MIN	WHAT
A	40	Your environment: VS Code, an agent, your project
B	30	Vibe coding and agentic engineering
C	15	The homework

Your environment

Form groups

- **Each row is a group — help each other**
- **Stuck? Raise your hand**

What you are putting together

WHAT	WHAT IT DOES
VS Code	Where you work — files, editor, terminal
A coding agent	Claude Code or Codex, inside VS Code — it works on your files
Your project	Your repository, on your own machine

The goal for today: your project is open, and the agent can tell you what it is.

1. VS Code

1. **code.visualstudio.com** → the big download button
2. **File → Open Folder** — always open a folder, never a single file
3. **Cmd+Shift+P · Ctrl+Shift+P** — the command palette

2. An agent inside VS Code

Cmd+Shift+X · Ctrl+Shift+X → search → Install → Sign in. One is enough.



Claude Code for VS Code

Claude Code for VS Code: Harness the power of Claude Code without leaving your IDE

 Anthropic

NEEDS A PAID CLAUDE PLAN — PRO OR ABOVE



Codex – OpenAI's coding agent

Codex is a coding agent that works with you everywhere you code — included in ChatGPT Plus

 OpenAI

NEEDS A CHATGPT ACCOUNT — THE FREE TIER IS LIMITED

3. Let the agent install the rest

An empty folder, **course**, in Documents → **File** → **Open Folder**

Then, to the agent:

```
Install git and Python if they are missing.  
Where possible, without an administrator password.  
Tell me which steps I have to do by hand.
```

What is left for you to do by hand, it tells you.

4. Your project, onto your machine

To the agent:

```
Clone https://github.com/esst-prog2/<your-repo> into this folder.
```

Then **File** → **Open Folder** → your project's folder.

*Your repository's name is on the Canvas page **Homework 1 — the projects**.*

If git will not install

1. Your repository on GitHub → green **Code** button → **Download ZIP**
2. Unzip it
3. **File** → **Open Folder** → that folder

Today's goal is still met. Git can wait for next week.

5. Look around

With your project open, to the agent:

Look around

It tells you about your project, from your README. **That is the check.**

Vibe coding and agentic engineering

“Vibe coding”

Andrej Karpathy · February 2025 · the post that named it

There's a new kind of coding I call “vibe coding”, where you fully give in to the vibes, embrace exponentials, and forget that the code even exists.

I “Accept All” always, I don't read the diffs anymore.

It's not too bad for throwaway weekend projects.

The same thing, named precisely

Simon Willison · October 2025

Vibe coding: the fast, loose and irresponsible way of building software with AI — entirely prompt-driven, and with no attention paid to how the code actually works.

He proposed a name for the other thing: **vibe engineering** — professionals who speed up with LLMs “*while staying proudly and confidently accountable for the software they produce.*”

Agentic engineering

Andrej Karpathy · Sequoia AI Ascent · 30 April 2026

Vibe coding raises the floor. Agentic engineering is about extrapolating the ceiling.

*Agentic engineering is about preserving the quality bar of professional software.
You are still responsible for your software, just as before.*

What the difference is

VIBE CODING

- You look at **what it does**
- The prompt is the whole plan
- An error message goes straight back in
- Done when it seems to work

AGENTIC ENGINEERING

- You also look at **how it does it**
- A spec, then a plan, then small steps
- A failing test says what is wrong
- Done when **you can show it works**

This course is about the second one

Agentic engineering.

Everyone takes away as much as they want.

A hint for using AI

The biggest problem is not that it hallucinates. It is that what it gives you is **irrelevant**.

RECOMMENDED METHOD

1. I think about the problem
2. I ask the AI for a **second opinion**
3. I decide — and move on

The homework

1. Snake — you are the tester

- Fork **esst-prog2/snake**
- Build it with the agent — **prompts only, do not open the code**
- **Play it.** Write down what is wrong
- Then fix it — now you read the code

2. Oware — the tests do not say everything

- Fork `esst-prog2/oware`
- Set it up, run the tests — all red
- Get them green with the agent
- Find the rules the tests do not check — test them, fix them
- Play a whole game against the computer

I will run more tests on your fork than the ones you have.

By Friday

1. **Snake** — your fork: the game, and a short note on what playing found
2. **Oware** — your fork: tests green, **your own tests** for the rules the given ones miss, and a way to play against the computer

Deadline: Friday 18 September, 20:00

Do not edit the tests you were given. Add your own next to them.

Exit ticket

1. Is your project open, and did the agent describe it? **yes / no / partly**
2. Where did your setup get stuck, if anywhere?
3. One sentence: what is the difference between vibe coding and agentic engineering?

Thank you